Introduction to Week 5

From Unstable Inversion to Ridge Regression

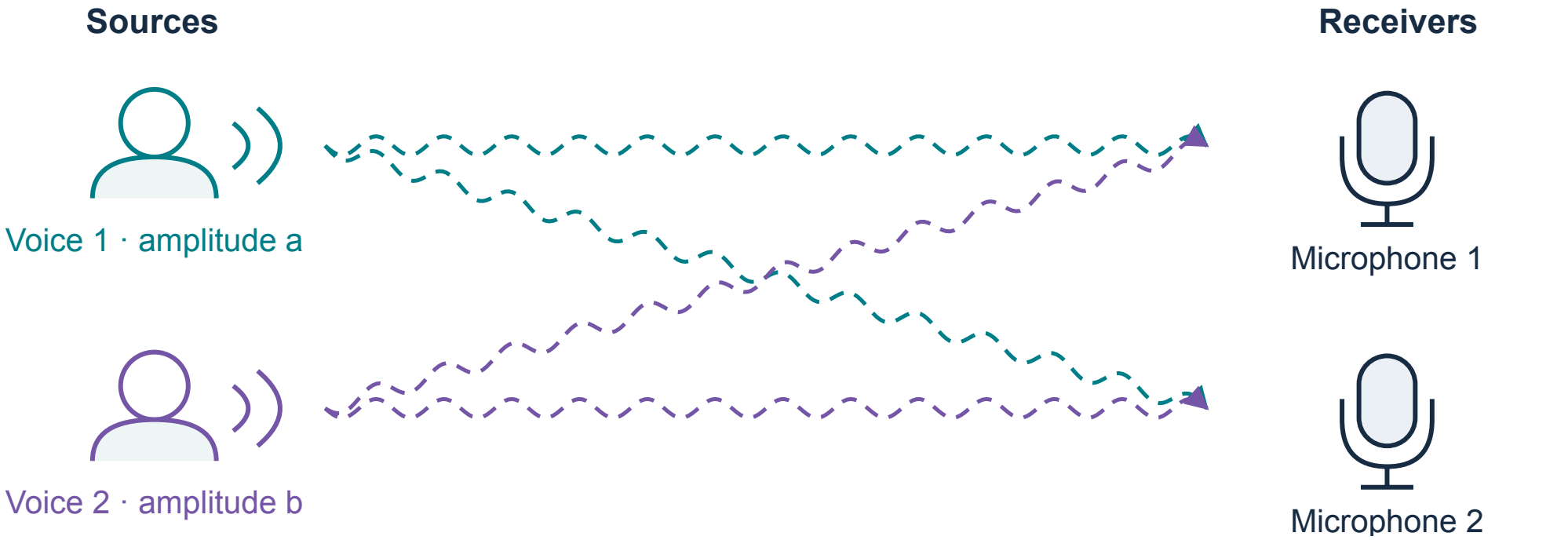
Bias–variance · inverse problems · Tikhonov regularization

DSAA3073 · Theories in Data Science

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Two voices, two microphones

Both microphones hear both voices. Can we recover the two voices from the recordings?



Unknown source amplitudes: $\mathbf{f} = (a, b)^{\mathrm{T}}$

Measured recordings: $\mathbf{y} = (y_1, y_2)^{\mathrm{T}}$

Original toy example · dashed waves indicate reception of each voice at each microphone

From recordings to a linear inverse problem

At one instant, a and b are the signed source amplitudes. The mixing weights are known, but the recordings contain noise.

$$\mathbf{y} = \mathbf{A}\mathbf{f} + \boldsymbol{\varepsilon}.$$

Two scalar measurement equations

$$y_1 = 0.51a + 0.49b + \varepsilon_1,$$
$$y_2 = 0.49a + 0.51b + \varepsilon_2.$$

Each microphone hears almost the same mixture.

One matrix equation

$$\mathbf{A} = \begin{pmatrix} 0.51 & 0.49 \\ 0.49 & 0.51 \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}.$$

Given $\mathbf{A}$ and $\mathbf{y}$, recover $\mathbf{f}$.

The actual measurement error $\boldsymbol{\varepsilon}$ is unknown.

Predict: would a 0.01 change in each recording produce only a small change in the separated voices?

A tiny recording error becomes a large source error

Suppose the true amplitudes are $\mathbf{f}^\star = (1, 0)^{\mathrm{T}}$.

Recording	Observed $\mathbf{y}$	Exact solution $\mathbf{A}^{-1}\mathbf{y}$
No measurement error	$(0.51, 0.49)^{\mathrm{T}}$	$(1, 0)^{\mathrm{T}}$
Add $(0.01, -0.01)^{\mathrm{T}}$	$(0.52, 0.48)^{\mathrm{T}}$	$(1.5, -0.5)^{\mathrm{T}}$

The sum is easy to observe

Noiseless: $y_1 + y_2 = a + b$.
The added noise cancels in this sum.

The difference is weakly observed

Noiseless: $y_1 - y_2 = 0.02(a - b)$.
Inversion multiplies difference noise by **50**.

💡 Agreement with noisy measurements does not guarantee agreement with the true sources.

Why the obvious fixes do not settle the problem

First idea	What remains unresolved
Solve with more precision	This error came from measurement noise. Exact arithmetic still amplifies it.
Use the pseudoinverse	For this invertible matrix it equals the inverse. Small positive singular values still give large gains.
Drop the difference entirely	Both voices share the sum. Discarding their difference also discards real separation information.
Always use stronger shrinkage	Shrinkage reduces coefficients toward zero. A larger penalty reduces noise amplification but can also suppress true signal, so expected error can rise.



Identify which errors come from the measurements and which directions amplify them. Then calculate how reducing amplification changes bias and variance.

Four ideas that apply beyond this example

1 · Separate error sources

Bias, variation across training samples, and noise in a new response need different explanations.

2 · Change coordinates

The SVD exposes directions that the observations determine strongly, weakly, or not at all.

3 · Change the objective

Penalize large coefficients to control amplification. Calculate the signal lost in return.

4 · Check what each result proves

Existence, uniqueness, stability, and prediction accuracy require different arguments.

Route: error decomposition → sensitive directions → Tikhonov and ridge → bias–variance tradeoff and limits.

The problem: recover a signal from noisy observations

$$\mathbf{y} = \mathbf{A}\mathbf{f}^\star + \boldsymbol{\varepsilon}, \quad \mathbf{A} \in \mathbb{R}^{n \times d}, \quad \mathbf{f}^\star \in \mathbb{R}^d.$$

What is given?

The known matrix $\mathbf{A}$ maps hidden source amplitudes to noiseless measurements; $\mathbf{y}$ is the observed data.
The hidden target $\mathbf{f}^\star$ and the measurement error are unknown.

What do we want?

A recovered signal or predictor that remains useful when observations change slightly.
Are we trying to recover the hidden parameters, or predict outputs accurately?

Regression is the same optimization pattern:

$$\mathbf{A} = \mathbf{X}, \quad \mathbf{f} = \boldsymbol{\beta}, \quad \mathbf{y} \approx \mathbf{X}\boldsymbol{\beta}.$$

Centering means subtracting each sample mean from the predictors and response. In the later moment calculation, the predictor matrix is fixed while response noise varies across repetitions; the intercept is unpenalized.

First separate two sources of randomness

At a fixed input x_0, let S be the random training sample; $\hat{f}_S(x_0)$ is the prediction fitted from S; $f^\star(x_0)$ is the true mean response; and $m(x_0)$ is the average fitted prediction over possible S. A new response is $Y = f^\star(x_0) + \varepsilon$, where ε is independent of S, has mean zero, and represents noise in that new response.

$$\mathbb{E}_{S,\varepsilon}\left[\left(Y - \hat{f}_S(x_0)\right)^2\right] = \underbrace{\left(m(x_0) - f^\star(x_0)\right)^2}_{\text{squared bias}} + \underbrace{\operatorname{Var}_S\left(\hat{f}_S(x_0)\right)}_{\text{variance across training samples}} + \underbrace{\sigma^2(x_0)}_{\text{noise in a new response}},$$

$m(x_0) = \mathbb{E}_S\left[\hat{f}_S(x_0)\right]$. The identity assumes squared loss and finite second moments.

Bias

The average fitted prediction differs from the true mean response.

Variance across training samples

The fitted prediction changes when the training sample changes.

Noise in a new response

A new response varies even when its mean is known.

The identity names the error components. Geometry will explain one mechanism of instability.

An inverse can exist and still be unreliable

Hadamard: three separate questions

1. Does a solution exist for every admissible observation?
2. Is that solution unique?
3. Does it depend continuously on the observation?

SVD: how much does each direction change?

$$\mathbf{A}\mathbf{v}_j = \sigma_j \mathbf{u}_j.$$

A change along source direction $\mathbf{v}_j$ produces an observation change scaled by σ_j.
If σ_j is small, that source change is hard to distinguish from measurement noise. Inversion divides by σ_j and amplifies the noise.
Zero: measurements do not detect that direction.



Our inverse exists and varies continuously, but its smallest singular value is 0.02. The difference-direction noise is multiplied by 50.

Tikhonov changes the objective; ridge specializes it

General finite-dimensional inverse

$$\mathbf{f}_\lambda = \operatorname*{argmin}_{\mathbf{f}} \left(\|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \lambda \|\mathbf{f}\|^2 \right).$$

Squared measurement error + penalty on coefficient size. A squared norm sums the squared components; $\lambda > 0$ controls penalty strength.

$$(\mathbf{A}^{\mathrm{T}}\mathbf{A} + \lambda \mathbf{I})\mathbf{f}_\lambda = \mathbf{A}^{\mathrm{T}}\mathbf{y}.$$

Centered linear regression: ridge

$$\hat{\boldsymbol{\beta}}_\lambda = \operatorname*{argmin}_{\boldsymbol{\beta}} \left(\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 \right).$$

$$\hat{\boldsymbol{\beta}}_\lambda = \left(\mathbf{X}^{\mathrm{T}}\mathbf{X} + \lambda \mathbf{I}\right)^{-1} \mathbf{X}^{\mathrm{T}}\mathbf{y}.$$

The same derivation with $\mathbf{A} = \mathbf{X}$.

The displayed residual term is a sum, with no $\frac{1}{n}$, and all displayed coefficients are penalized. With an averaged residual convention, $\lambda = n\lambda'$ gives the same estimator.

💡 The penalty changes which solution we prefer. It does not remove the measurement noise.

The mechanism: damp weak directions more strongly

Data-to-solution gain

$$\frac{1}{\sigma_j} \quad \rightarrow \quad \frac{\sigma_j}{\sigma_j^2 + \lambda}.$$

Acts on the measured coordinate $\mathbf{u}_j^{\mathrm{T}}\mathbf{y}$ to produce an estimated source coordinate.

Fraction of the ordinary least-squares (OLS) coefficient retained

$$s_j(\lambda) = \frac{\mu_j}{\mu_j + \lambda}, \quad \mu_j = \sigma_j^2 > 0.$$

This compares the ridge coefficient with the OLS coefficient in the same direction. In regression, μ_j is the corresponding eigenvalue of $\mathbf{X}^{\mathrm{T}}\mathbf{X}$: a squared singular value.

For the microphones, set $\lambda = 0.0004$:

Direction	Exact gain	Ridge gain	Fraction kept
Sum: $\sigma = 1$	1	0.9996	99.96%
Difference: $\sigma = 0.02$	50	25	50%

Ridge reduces both noise amplification and true signal in the weak direction.

What we can prove before testing prediction quality

A unique finite-dimensional solution

For every nonzero $\mathbf{z}$,

$$\mathbf{z}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\mathbf{A} + \lambda\mathbf{I})\mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \lambda\|\mathbf{z}\|^2 > 0.$$

The Hessian is positive definite. Normal equations give a solution; strict convexity shows no other minimizer exists.

An explicit stability bound

For fixed $\lambda > 0$, let $\Delta\mathbf{y}$ be a change in the observed measurements:

$$\|\mathbf{f}_\lambda(\mathbf{y} + \Delta\mathbf{y}) - \mathbf{f}_\lambda(\mathbf{y})\| \le \frac{\|\Delta\mathbf{y}\|}{2\sqrt{\lambda}}.$$

At $\lambda = 0.0004$, amplification is at most **25**, even with zero singular values.

💡 A unique, stable estimate can still have large prediction error.

A little bias can buy a useful variance reduction

In one weak direction, $t = 0.02\theta + \eta$: t measures source coefficient θ with measurement noise η. For this calculation, $\theta = 1$ and η has mean 0 and variance 0.0004.

$$\hat{\theta}_\lambda = \frac{0.02}{0.0004 + \lambda} t, \quad s = \frac{0.0004}{0.0004 + \lambda}.$$

Coefficient MSE = squared bias $(1 - s)^2$ + variance s^2.

Penalty λ	Fraction s	Squared bias	Variance	MSE
0	1	0	1	1
0.0004	0.5	0.25	0.25	**0.50**
0.0036	0.1	0.81	0.01	0.82

The table measures coefficient error. Prediction error measures errors in predicted outputs; predicting a new response also includes its noise variance.



A moderate penalty improves this example. Stronger shrinkage reduces variance further but can make total error worse.

Constructed weak-direction calculation · moment formulas: learning sheet PDF pp. 24–27

Three limits to keep in view

1. **Missing information stays missing.** A null direction is a coefficient change that produces no change in the measurements. The squared-norm penalty chooses zero for a coefficient the data do not determine; it does not recover that coefficient.
2. **Noise in a new response remains.** A penalty cannot remove response variance left even when the conditional mean is known.
3. **The best penalty is still unknown.** Training residuals do not select the best predictive tradeoff. Later selection needs independent evidence or another justified criterion.

As $\lambda \to 0^+$, finite-dimensional ridge approaches the least-squares solution with the smallest Euclidean norm. It equals the usual OLS inverse formula only under full column rank.

Six distinctions that beginners often find hard

Hard distinction	Question to ask yourself
Training randomness / noise in a new response	What is being redrawn, and what stays fixed inside this expectation?
Zero / small singular value	Is information absent, or present but easily overwhelmed?
Gain / shrinkage fraction	Does this factor multiply a measured coordinate, a fitted coordinate, or an OLS coefficient?
Coefficient / prediction error	Which directions does the test input emphasize? Where does noise in a new response enter?
Unique / accurate solution	Did the proof establish uniqueness, sensitivity to measurement changes, or expected prediction accuracy?
Fixed penalty / selected penalty	Is λ fixed in the theorem, or chosen after looking at data?

For each formula, identify what is fixed, what is random, and which error is being measured.

Connections to the undergraduate mathematical core

Course / topic	Exact bridge	Why it helps here
Linear algebra	$\mathbf{A}\mathbf{v}_j = \sigma_j \mathbf{u}_j$; $\mu_j = \sigma_j^2$ for $\mathbf{X}^{\mathrm{T}}\mathbf{X}$	Turn a coupled matrix problem into scalar directions.
Numerical analysis	Inverse norm, condition number, measurement error versus rounding error	Measure how much input error can change the answer.
Probability and statistics	Conditional expectation; centered cross terms; covariance transforms	Derive bias and variance without mixing random sources.
Multivariable calculus and convex optimization	Gradient, Hessian, positive definiteness; norm budgets and penalties	Derive the estimator and prove that it is the unique minimum.

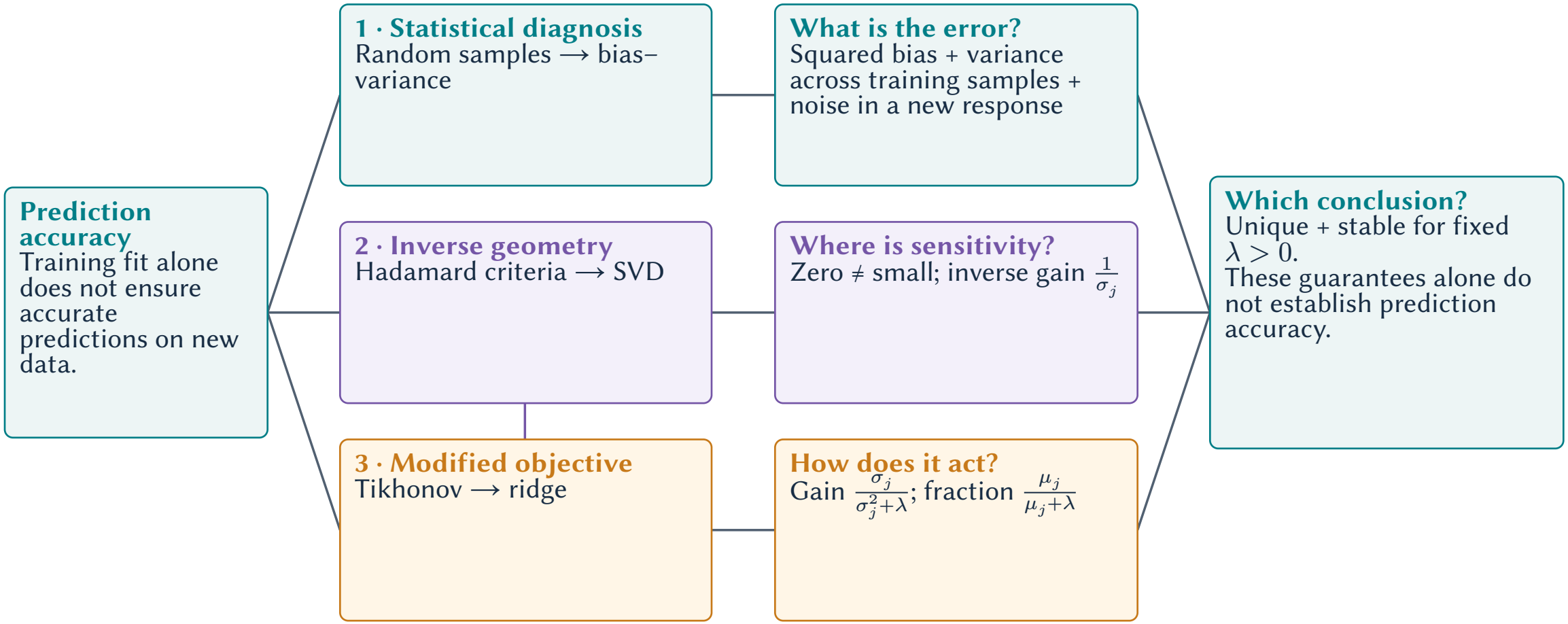
The constrained-optimization/KKT correspondence is an optional extension; see the sheet's PDF p. 15.

Connections to other CS, AI, and DS courses

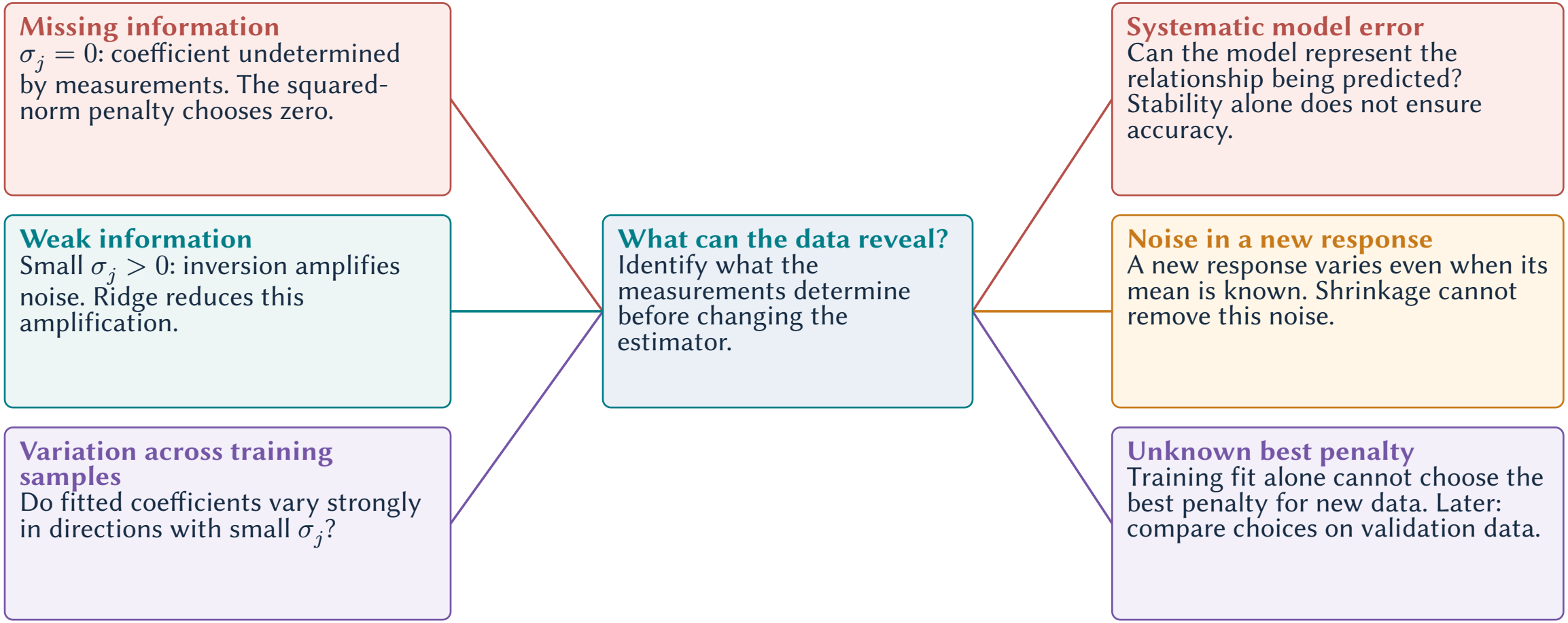
Neighboring topic	Connection worth carrying forward
Signal processing / computational imaging	Deblurring and source separation recover changes weakened by the measurement process. Ridge reduces amplification in each direction.
Machine learning theory	Norm-bounded linear classes illustrate fitting data while controlling coefficient size. A penalty alone does not change the VC dimension of an unchanged class.
Experimental design / sensing	Change the measurements to make weak directions more informative. This complements changing the estimator.
Model evaluation	Prediction risk means expected prediction error on new data. Selecting a penalty and evaluating the selected procedure need distinct evidence; keep the final test set untouched.

Optional Bayesian connection: with Gaussian observation noise and an isotropic, zero-mean Gaussian prior, ridge gives the maximum a posteriori (MAP) estimate—the parameter value with highest posterior density.
These are conceptual links, not extra assessed prerequisites. Parameter-selection algorithms belong to the following chapter.

Mindmap 1 · The academic progression



Mindmap 2 · What can the data actually reveal?



Return to the microphones: what should we do?

What we know

- The difference direction is weakly observed.
- Exact inversion amplifies its measurement noise by 50.
- Ridge reduces noise amplification but also shrinks the true source coefficients, introducing bias.

What we must still establish

- How large is the noise relative to the weak signal?
- Do we care about individual sources or only their sum?
- What independent evidence supports the penalty choice?



The week's calculations show how a penalty changes bias, variance, and total expected error.